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A APPENDIX: PROOF OF PROPOSITIONS

A.1 Proposition 1

To prove Proposition 1, we restate a result from [23] for a MDP. Here γ is the discount factor, t refers to steps of the optimization, and $d_\mu^\pi(\gamma) = \mathbf{E}_{s_0\sim\mu}(1-\gamma)\sum_{t=0}^{\infty}\gamma^t P(s_t = s|s_0,\pi)$ is the discounted state distribution.

Theorem 2 ([23]). *Let Assumption 1 hold and let $\{\theta_t\}_{t\geq 1}$ be generated using $\theta_{t+1} \leftarrow \theta_t + \eta\frac{\partial V^{\pi_{\theta_t}}(\mu)}{\partial\theta_t}$ where $V^{\pi_{\theta_t}}(\mu) = \mathbf{E}_{s\sim\mu}V^{\pi_{\theta_t}}(s)$. Let $\eta = (1-\gamma)3/8$, c be the positive constant $c := \inf_{s\in S, t\geq 1}\pi_{\theta_t}(a^*(s)|s) > 0$. Then, for all $t \geq 1$,*

$$\mathbf{E}_{s\sim\mu}[V^*(s) - V^{\pi_{\theta_t}}(s)] \leq \frac{16S}{c^2(1-\gamma)^6 t}\left\|\frac{d_\mu^{\pi^*}(\gamma)}{\mu}\right\|_\infty^2 \left\|\frac{1}{\mu}\right\|_\infty \tag{10}$$

Proposition 1. *Let Assumptions 1 and 2 hold and let $\{\theta_t\}_{t\geq 1}$ be generated using the standard policy gradient update: $\theta_{t+1} \leftarrow \theta_t + \eta\frac{\partial V^{\pi_{\theta_t}}(\mu)}{\partial\theta_t}$ where $V^{\pi_{\theta_t}}(\mu) = \mathbf{E}_{x\sim\mathcal{D}}\mathbf{E}_{\mathbf{y}\sim\pi_{\theta_t}(.|x)}\mathcal{R}(x,\mathbf{y})$. Then, for all $t \geq 1$, with $\eta = 1/8$,*

$$\mathbf{E}_{x\sim\mathcal{D}}[(\pi^*(x) - \pi_{\theta_t}(x))^T\mathbf{r}(x)] \leq \frac{16SA^2\rho^2}{t}\left\|\frac{1}{\mu}\right\|_\infty \tag{5}$$

where $S = |X|$ is the number of states and A is the number of actions.

Proof. Note that our RL setup is one-step, hence we can assume $\gamma = 0$. Then $V^\pi(s)$ from Theorem 1 simplifies to,

$$V^\pi(s) = \mathbf{E}_{a_t\sim\pi(.|s_t)}\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t) = \mathbf{E}_{a_t\sim\pi(.|s_t)}r(s_0,a_0) = \pi(s)^T\mathbf{r}(s)$$

where π and r in the last equation refer to the vectors of action probabilities and their rewards given a state. Further, since $\gamma = 0$, $d_\mu^{\pi^*}(\gamma) = \mu$. Hence, we can write Theorem 1 as,

$$\begin{aligned}
\mathbf{E}_{s\sim\mu}[V^*(s) - V^{\pi_{\theta_t}}(s)] &= \mathbf{E}_{s\sim\mu}[\pi^*(s)^T\mathbf{r}(s) - \pi_{\theta_t}(s)^T\mathbf{r}(s)] && (11)\\
&= \mathbf{E}_{s\sim\mu}[(\pi^*(s) - \pi_{\theta_t}(s))^T\mathbf{r}(s)] && (12)\\
&= \mathbf{E}_{x\sim\mathcal{D}}[(\pi^*(x) - \pi_{\theta_t}(x))^T\mathbf{r}(x)] && (13)\\
&\leq \frac{16S}{c^2(1-\gamma)^6 t}\left\|\frac{d_\mu^{\pi^*}(\gamma)}{\mu}\right\|_\infty^2\left\|\frac{1}{\mu}\right\|_\infty && (14)\\
&\leq \frac{16S}{c^2 t}\left\|\frac{1}{\mu}\right\|_\infty && (15)
\end{aligned}$$

where the third equality is because the initial state distribution is the distribution of queries in the training data.

Now, using Assumption 2, the minimum initial probability for the optimal action a^* is $\frac{1}{\rho A}$ for all states. Assuming that the gradient updates do not decrease the probability of the optimal action, $c = \inf_{s\in S, t\geq 1}\pi_{\theta_t}(a^*(s)|s) = \frac{1}{\rho A}$. Substituting c in the above equation, we obtain the result.

$$\mathbf{E}_{x\sim\mathcal{D}}[(\pi^*(x) - \pi_{\theta_t}(x))^T\mathbf{r}(x)] \leq \frac{16SA^2\rho^2}{t}\left\|\frac{1}{\mu}\right\|_\infty$$

□

A.2 Proposition 2

Proposition 2. *With the new formulation, under the assumptions of Proposition 1, for all $t \geq 1$,*

$$\mathbf{E}_{x\sim\mathcal{D}}(\pi^*(x) - \pi_{\theta_t}(x))^T\mathbf{r}(x) \leq \frac{64SA\rho^2}{t}\left\|\frac{1}{\mu}\right\|_\infty \tag{9}$$

Proof. Using the equation from Proposition 1 and substituting $S = SA$ and $A = 2$ leads us to the result. □